

DATA ANALYTICS PORTFOLIO

Retail Analysis Project

Measuring the Impact of Diwali & Sankranti Campaigns Across 50 Stores in India

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Business Context



The Situation

Atliq ran two major promotional campaigns across 10 Indian cities. Each campaign tested five promotion types across 15 products and 50 stores.

Business Question: Which campaign, promotion, city, store, and product delivered the strongest growth — and where should the next budget go?



The Data

Table	Rows	Granularity
Products	15	Product-level
Stores	50	Store-level
Campaigns	4	Campaign-level
Events	10,150	Sales-level

Before/after promotional sales and quantity are assumed to cover comparable time periods.

Objectives

Five questions the analysis had to answer:

Campaign Impact

Measure sales and quantity growth before vs. after each campaign.

Regional Performance

Identify the best and worst performing cities and stores.

Category & Product

Rank categories and products by growth contribution.

Promo Effectiveness

Compare five promo types to find the highest ROI.

Diagnose Underperformance

Understand why some stores and promos fell below expectations.

Recommendations

Translate insights into actionable next-quarter strategy.

Data Model

Relationships

A star-schema-inspired model linking events (fact) with products, stores, and campaigns (dimensions).

Fact table: events — 10,150 sales records with before/after promo data.

Dimensions: products, stores, campaigns.

Data Preparation Steps

- ▶ Verified relationships between all tables
- ▶ Confirmed no missing values in any table
- ▶ Created calculated columns for before/after sales
- ▶ Built DAX measures for Sales Growth % and Qty Growth %

Key DAX Measures

```
Sales Growth % =  
DIVIDE(  
    SUM(events[Sales After Promo]) - SUM(events[Sales Before Promo]),  
    SUM(events[Sales Before Promo])  
    ) * 100
```

```
Qty Growth % =  
DIVIDE(  
    SUM(events[Qty After Promo]) - SUM(events[Qty Before Promo]),  
    SUM(events[Qty Before Promo])  
    ) * 100
```

Campaigns & Promotion Types

Two Campaigns

Diwali
12 – 18 November 2023

Sankranti
10 – 16 January 2024

Five Promotion Types

Promo	Type
25% OFF	Percentage discount
33% OFF	Percentage discount
50% OFF	Percentage discount
₹500 Cashback	Fixed-value incentive
BOGOF	Buy One Get One Free

Campaign Growth — Headline

Diwali dramatically outperformed Sankranti on both sales and quantity.

Sales Growth

94.12%

DIWALI · ₹160.29M

50.87%

SANKRANTI · ₹87.7M

Quantity Growth

155.31%

DIWALI · 252,069 UNITS

66.25%

SANKRANTI · 183,404 UNITS

Takeaway: Diwali nearly doubled Sankranti's sales growth and more than doubled its quantity growth — clear evidence the Diwali promotion captured stronger consumer intent.

City-Wise Growth

Top 5 Cities — Sales Growth

City	Sales Growth
Madurai	86.81%
Bengaluru	81.18%
Chennai	80.52%
Vijayawada	79.58%
Mysuru	77.79%

Top 5 Cities — Qty Growth

City	Qty Growth
Madurai	115.58%
Coimbatore	114.33%
Bengaluru	113.83%
Chennai	110.79%
Trivandrum	110.43%

Insight: Madurai leads on both metrics. Visakhapatnam is the only city below 100% quantity growth (97.47%) — a candidate for store-level diagnostics.

Store Performance

Top 5 Sales Growth Stores

Store	Growth
STMIDU-0	105.64%
STMYS-3	104.34%
STBLR-0	103.11%
STMYS-1	102.68%
STCHE-7	102.48%

Bottom 5 Sales Growth Stores

Store	Growth
STMYS-0	39.43%
STCHE-1	45.00%
STHYD-1	45.92%
STVSK-3	46.04%
STMILR-0	48.09%

24 stores grew slower than their city average — a pattern that suggests operational rather than market-wide issues (footfall, service, or local execution).

Promotion Type Performance

Promo Type	Sales Growth	Qty Growth	Verdict
₹500 Cashback	136.11%	183.33%	★ Best for revenue
BOGOF	83.68%	269.98%	★ Best for volume
33% OFF	-4.28%	43.04%	Weak
50% OFF	-33.60%	32.63%	Poor
25% OFF	-34.61%	-12.99%	✖ Underperforming

Key insight: Discount depth is not the driver of growth. The two *non-discount* mechanics — Cashback and BOGOF — delivered the strongest results. Deep discounts like 50% OFF and 25% OFF actually *reduced* sales.

Category Growth

Category	Sales Growth	Qty Growth
Combo1	136.11%	183.33%
Home Appliances	82.61%	264.39%
Home Care	46.83%	106.60%
Grocery & Staples	9.61%	94.02%
Personal Care	-34.20%	24.50%

Watch-out: Personal Care is the only category with negative sales growth (-34.20%) and it's driven entirely by the failed 25% OFF and 50% OFF promos. Rework its promotional strategy.

Product Highlights

Top Revenue Products

Product	Sales Growth
Atliq Home Essential 8-Product Combo	136.11%
Waterproof Immersion Rod	83.09%
High Glo 15W LED Bulb	81.49%
Double Bedsheet Set	79.13%
Curtains	77.67%

Bottom Products

Product	Sales Growth
Body Milk Nourishing Lotion	-35.95%
Fusion Container Set of 3	-35.59%
Scrub Sponge for Dishwash	-35.11%
Cream Beauty Bathing Soap	-35.00%
Lime Cool Bathing Bar	-33.40%

Four of the bottom five are **Personal Care** products — confirming the category-level finding.

Key Insights

- 1 **Diwali dominated Sankranti** — 94% vs 51% sales growth; 155% vs 66% quantity growth.
- 2 **Cashback > discount** — ₹500 Cashback (136% sales) and BOGOF (270% qty) beat every percentage discount.
- 3 **Madurai was the standout city** — highest growth on both sales and quantity.
- 4 **STMDU-0 led sales growth**, while STMYS-1 led quantity growth.
- 5 **Combo1 was the top category** — 136% sales growth, 183% quantity growth.
- 6 **25% OFF underperformed everywhere** — negative growth across all categories and campaigns.
- 7 **Personal Care needs a reboot** — the only category with negative sales growth (-34.20%).

Business Recommendations

Budget Reallocation

Shift spend toward volume-driven mechanics (Cashback, BOGOF) and phase out 25% OFF.

Store-Level Review

Audit the 24 stores below their city average for footfall, service, staffing, and local execution.

Category Strategy

Rework Personal Care promotions; add SKUs to high-growth categories like Combo and Home Appliances.

Marketing Collaboration

Partner with marketing to redesign low-performing promos and support weak stores with targeted campaigns.

Customer Loyalty

Segment top customers and pilot a loyalty program to lock in repeat purchase behaviour.

Geographic Expansion

Evaluate new store locations in high-growth cities such as Madurai, Bengaluru, and Chennai.

Thank You

Questions & Discussion

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